Is HLE Measurement-Valid?

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1 Overview and motivation

Humanity's Last Exam (HLE) has quickly become a standard instrument for evaluating frontier language models, appearing in capability assessments and policy discussions shortly after its release. Designed to resist retrieval and pattern matching through expert-level questions across mathematics, natural sciences, and humanities, it is a reasonable candidate for tracking genuine progress at the frontier. However, widespread adoption has outpaced any serious scrutiny of its measurement properties. Most benchmark papers, HLE included, report aggregate accuracy and domain-specific subscores without testing whether those subscores reflect anything real. They appear on leaderboards and get interpreted as evidence that one model outperforms another at mathematics or chemistry, an interpretation that assumes domain categories correspond to empirically distinct latent capabilities rather than a single general reasoning factor.

This paper addresses those gaps directly. We evaluate 40 language models on the text-only multiple-choice subset of HLE and apply psychometric methods drawn from the course framework [1] to ask three questions: (a) **Which items actually discriminate between models?**, addressed via 2PL Item Response Theory (Chapter 2); (b) **Whether HLE's subject-domain structure recovers as distinct latent constructs or collapses to a general factor?**, addressed via confirmatory factor analysis and construct validity tools (Chapters 2 and 6); and (c) **Whether the benchmark measures the same construct equivalently across reasoning-specialized and standard instruction-tuned models?**, addressed via measurement invariance testing, connecting to differential item functioning and distribution shift (Chapters 6 and 7). We replicate core analyses on GPQA Diamond to assess whether findings reflect genuine properties of frontier benchmarks or artifacts of HLE's construction.

2 Literature Review

Benchmarks are the primary instrument by which frontier model progress is measured, but their useful lifespan is short. For instance, popular benchmarks such as MMLU have saturated within a few years of release [2]. This has motivated the development of HLE: a benchmark of 2,500 expert-level questions across mathematics, humanities, and the natural sciences, crowdsourced from nearly 1,000 subject-matter experts across 50 countries and filtered to ensure that frontier LLMs could not answer them correctly at submission time [2]. HLE and other frontier benchmark studies (e.g., MMLU [3], GPQA [4]) typically report aggregate accuracy as their primary evaluation metric. However, such metrics obscure item-level measurement properties, including whether items discriminate between models, whether domain subscores reflect distinct latent constructs, and whether scores are comparable across model families. These properties matter because benchmark rankings increasingly inform high-stakes decisions: from model selection in deployment to AI governance and policy [5]. When leaderboards place reasoning-specialized and standard instruction-tuned models on the same scale, or when domain subscores are interpreted as evidence of distinct capabilities, the validity of those comparisons is assumed rather than demonstrated. Decisions made on the basis of invalid measurements risk systematically misrepresenting which models are most capable and where progress is actually occurring [6].

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Psychometric approaches offer a framework for evaluating these measurement properties directly. Applying IRT across 29 NLP datasets, Vania et al. [7] find that several widely used benchmarks contain items that fail to discriminate between strong models, raising fundamental questions about whether aggregate scores on such benchmarks reflect genuine differences in capability at all. Leaderboards routinely report and interpret domain-specific subscores, for example, treating "math" or "chemistry" performance as separable capabilities, without testing whether these categories recover as empirically distinct latent factors rather than indicators of a single underlying reasoning ability.

A small but growing literature has begun applying psychometric methods to LLM benchmarks. TinyBenchmarks [8] uses IRT to enable efficient evaluation on subsets of MMLU and related benchmarks, while Chatbot Arena [9] applies Bradley-Terry models to pairwise model comparisons. These studies demonstrate the feasibility of treating benchmarks as measurement instruments, but they focus primarily on item difficulty estimation or ranking efficiency rather than on dimensionality and invariance questions that determine whether domain-specific capability claims are justified.

A related and largely unexamined assumption concerns the comparability of scores across reasoning-specialized and standard instruction-tuned models. Leaderboards, including HLE's own, place these model families on the same scale without formally testing whether the benchmark measures the same construct equivalently across groups. To our knowledge, no such psychometric analysis has been applied to HLE. **This study addresses that gap by estimating item-level discrimination and difficulty under a 2PL IRT model, testing whether HLE's eight subject domains recover as distinct latent factors, and assessing whether scores are comparable across reasoning-specialized and standard instruction-tuned models.**

3 Proposed Methods and Analysis

3.1 Dataset: HLE Subset

The full HLE benchmark contains two question formats: exact-match short-answer (≈76%) and multiple-choice with five or more answer choices (≈24%). Approximately 14% of all questions require image comprehension. Subject coverage skews toward quantitative domains: mathematics (41%), biology/medicine (11%), computer science/AI (10%), physics (9%), humanities/social science (9%), other (9%), chemistry (7%), and engineering (4%). We restrict to the *text-only multiple-choice* subset, which is the only subset amenable to automated binary scoring without multimodal infrastructure or LLM-judge extraction (required for `exactMatch` items). This subset comprises approximately $J \approx 516$ items (24% of the ≈2,150 text-only questions). The full dataset is publicly available via HuggingFace (`cais/hle`, `split="test"`) and is filtered to this subset by selecting `answer_type == "multipleChoice"` and excluding image-dependent items.

◇ **Example Items from HLE**

Analytic Subset. To illustrate the difficulty and style of `multipleChoice` items in our text-only analytic subset:

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1. **Philosophy:** "Which condition of Arrhenius's sixth impossibility theorem do critical-level views violate?" with answer choices including Egalitarian Dominance, General Non-Extreme Priority, Non-Elitism, Weak Non-Sadism, and Weak Quality Addition.

2. **Linguistics:** "What is the standard Japanese pitch accent pattern of the word for 'younger brother' in Japanese?" with choices spanning Heiban, Atamadaka, Nakadaka, Odaka, and Heiban/Nakadaka.

Note: These items demonstrate the core challenge of HLE: they cannot be answered by retrieval or pattern matching and require genuine domain expertise, making them valuable for IRT-based analysis.

3.2 Response Matrix Construction

We will construct a binary response matrix $\mathbf{X} \in \{0,1\}^{N \times J}$ where $N \approx 40$ models and $J \approx 516$ items (Chapter 1). Items with zero variance across all models (i.e., answered correctly or incorrectly by all models) will be excluded prior to fitting, as they carry no discriminative measurement information.

Model list. We will include approximately 40 models, partitioned into two groups, with the final list determined by data availability. A model is classified as *reasoning-specialized* if it uses extended chain-of-thought, process reward models, or test-time compute scaling as a core part of its inference procedure (e.g., o3, o3-pro, DeepSeek-R1, Gemini 2.5 Pro Thinking). The second group comprises *standard instruction-tuned* models (e.g., GPT-4o, GPT-4.5, Claude 3.5 Sonnet, Llama 3.1 405B).

Response collection. For models with publicly available item-level HLE results (e.g., those on the Scale AI leaderboard), we use published results directly. For remaining models, we collect responses via each provider's API using the standard HLE prompt template. Since we restrict to multiple-choice items, responses are binarized by direct string matching against the ground-truth answer letter.

3.3 Analyses

Item Response Theory (2PL). We plan to use the 2-PL model for an *estimation* problem: the target is the latent parameter vector $\{a_j, b_j, \theta_i\}$. The 2PL specifies (Chapter 2):

$$P(X_{ij} = 1 \mid \theta_i) = \sigma\big(a_j(\theta_i - b_j)\big), \tag{1}$$

where θ_i is model i's latent ability, b_j is item difficulty, and $a_j \geq 0$ is item discrimination. Parameters will be estimated via MML, fixing $\theta \sim \mathcal{N}(0, 1)$ for identification; we report 95% CIs throughout given the small N. We report the joint distribution of $(\hat{b}_j, \hat{a}_j)$ by domain, flag items with $\hat{a}_j \leq 0$, and examine item information functions $I_j(\theta) = a_j^2 P_j(\theta)(1 - P_j(\theta))$ and the test information function $I(\theta) = \sum_j I_j(\theta)$ to assess where HLE concentrates measurement precision along the ability continuum. IRT-based ability rankings are compared to raw accuracy rankings via Spearman ρ, and marginal reliability of $\hat{\theta}$ is reported to characterize the precision of model ability estimates.

Confirmatory Factor Analysis. Connecting to the factor models and structural validity framework studied, we test whether HLE's eight subject-domain labels correspond to empirically distinct latent capabilities within the text-only multiple-choice subset. Because this subset is not a random sample of HLE (filtering by format and modality may alter the domain composition relative to the full benchmark), conclusions are scoped to this subset; we report the domain distribution of included items to characterize any imbalance. We fit three models to the inter-item tetrachoric correlation matrix: a single general factor (M_1), eight domain-specific factors with items loading only on their designated factor (M_8), and a bifactor model (M_{bi}) allowing both. Given $N = 40$, fit statistics are treated as approximate guides and factor loadings are reported with bootstrapped standard errors. We report CFI, TLI, RMSEA, and SRMR for each model and compare M_1 vs. M_8 via the WLSMV-corrected $\Delta\chi^2$ test [10]; if $\Delta\text{CFI} < .010$, we conclude the domain labels do not recover as distinct constructs within this subset. We additionally report ω_h from M_{bi} to quantify variance attributable to the general factor vs. domain-specific factors.

Measurement Invariance. Connecting to DIF and distribution shift, we test whether HLE measures the same construct on the same scale for reasoning-specialized vs. standard instruction-tuned models. We fit the standard sequence of nested multi-group CFA models (configural $\rightarrow$ metric $\rightarrow$ scalar $\rightarrow$ strict), as suggested by Widaman & Reise [11]. Invariance at each step is evaluated using ΔCFI, ΔRMSEA, and ΔSRMR, with thresholds recommended by Chen [12] and Cheung & Rensvold [13]: $\Delta\text{CFI} < .010$, $\Delta\text{RMSEA} < .015$, and $\Delta\text{SRMR} < .030$ for metric invariance or $\Delta\text{SRMR} < .015$ for scalar and residual invariance. Scalar invariance is the critical threshold: failure implies that leaderboard comparisons across model families conflate true ability differences with measurement artifacts. As a secondary metric, we quantify the magnitude of $\hat{\theta}$ bias incurred when full invariance is incorrectly assumed, by comparing ability estimates under partial vs. full invariance constraints.

Robustness Analysis. To assess whether our findings reflect genuine properties of frontier capability benchmarks or are artifacts of HLE's construction, we replicate the core analyses (2PL IRT, M_1 vs.

M_8 CFA, measurement invariance) on GPQA Diamond [4], a 448-item graduate-level multiple-choice STEM benchmark sharing HLE's design intent but constructed entirely independently. Convergence across benchmarks would support the generalizability of our conclusions; divergence would suggest HLE-specific artifacts.

4 Timeline and Responsibilities

We divide the work into three phases across the weeks. Table 1 summarizes these weekly phases, responsibilities, and identified risks with mitigation strategies.

Table 1: Timeline, responsibilities, and risk mitigation.

Week	**Milestone**	**Responsible**	**Risk & Mitigation**
05/11–05/15	Data collection and API runs Response matrix verification Analysis environment setup	Mayank Savira Tyler	API access/cost: prioritize Scale AI leaderboard models; substitute inaccessible models while maintaining group balance
05/18–05/21	2PL IRT estimation Confirmatory factor analysis Measurement invariance testing GPQA Diamond replication	Mayank Savira Tyler Mayank	Small $N = 40$: report 95% CIs for all item parameters
05/22–05/27	Paper writing Figures and tables Related work and references Review, code compilation and submission	Savira Tyler Mayank Savira	CFA convergence: apply item-level variance filters; report bootstrapped standard errors on factor loadings

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